

PSG COLLEGE OF TECHNOLOGY,COIMBATORE-641004

23D501- IMAGE PROCESSING

MINI PROJECT

TOPIC:BLOOD CELL SUBTYPE IDENTIFIER

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BLOOD CELL SUBTYPE IDENTIFIER

Project Report

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Abstract

Blood cell analysis is an important part of medical diagnosis. Manual examination of microscopic blood-smear images can require significant time and expert observation. This project presents an automated image-based system for identifying four major white blood cell (WBC) subtypes: eosinophils, lymphocytes, monocytes, and neutrophils. The system uses image processing and deep learning to identify cell-like regions and predict their subtype. A pretrained DenseNet121 model is used to extract meaningful visual features from blood-cell images. The intended output is a labelled image showing detected cells and their predicted subtype, together with the number of detected cells.

1. Introduction

Blood cell analysis is an important part of medical diagnosis. Microscopic blood-smear examination is commonly used to observe blood-cell morphology. Different WBC subtypes show different visual characteristics, including differences in cell size, nucleus shape, and granules. The project described in the supplied presentation applies image processing and deep learning to automate WBC subtype identification from microscopic images. The system is designed to detect cell-like regions and predict the subtype of each detected region.

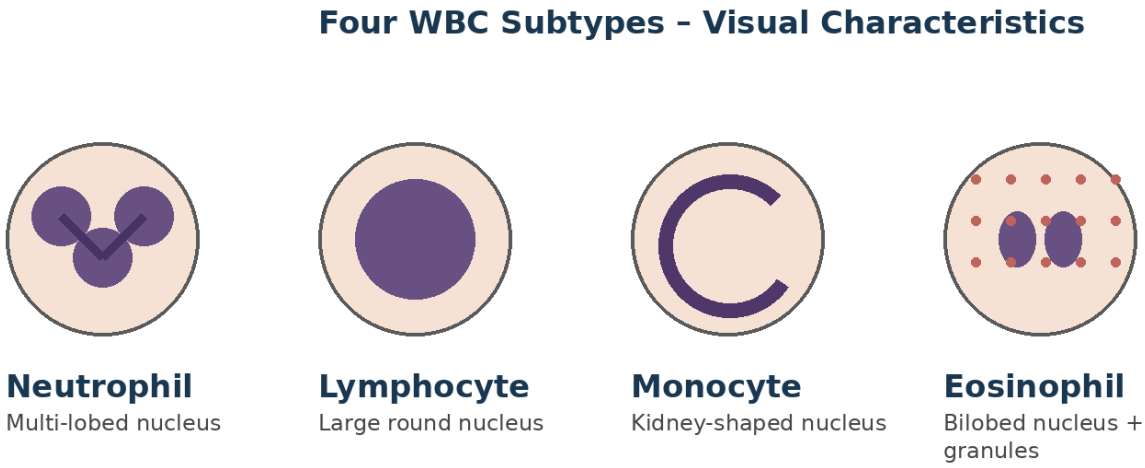


Figure 1. Simplified visual characteristics of the four WBC subtypes.

2. Problem Statement

Manual identification of WBC subtypes from microscopic images can be time-consuming and depends on expert observation. The project addresses the need for an automated image-based system that can detect cell-like regions, identify WBC subtypes, classify cells into four major subtypes, provide the number of detected cells, and display detected cells with their predicted labels.

3. Objectives

- Develop an automated computer-vision and deep-learning system for identifying WBC subtypes.
- Prepare and process the blood-cell image dataset.
- Extract meaningful visual features from blood-cell images.
- Classify WBCs into eosinophil, lymphocyte, monocyte, and neutrophil subtypes.
- Detect individual cell-like regions in an uploaded image.
- Predict the subtype of each detected region.

4. Methodology

The methodology consists of dataset preparation, feature extraction, and WBC subtype classification and detection. First, the blood-cell dataset is loaded, separated into training and testing images, and prepared for processing. Next, a pretrained DenseNet121 model is used to extract important visual features from the blood-cell images. These extracted features are used by the classification stage to distinguish the four target WBC subtypes. The system then detects cell-like regions and predicts the subtype of each detected region.

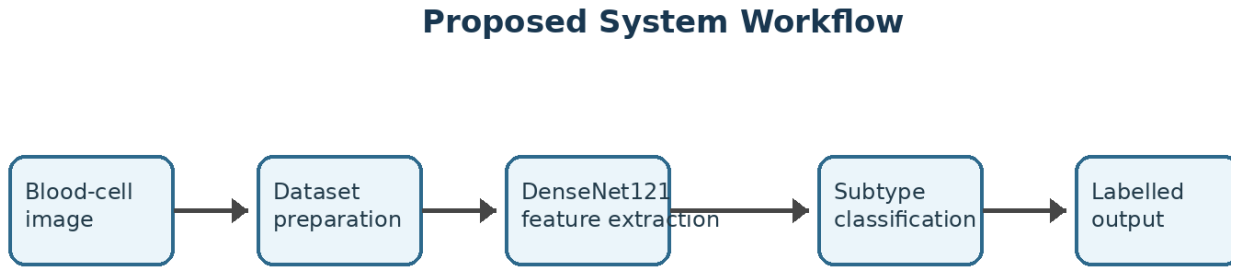


Figure 2. Simplified workflow of the proposed system.

5. WBC Subtype Characteristics

Subtype	Characteristic appearance	Key visual feature
Neutrophil	Multi-lobed nucleus with fine granules	Multi-lobed nucleus
Lymphocyte	Large round nucleus with a small rim of cytoplasm	Large round nucleus

Monocyte	Relatively large cell with abundant cytoplasm	Kidney/bean-shaped nucleus
Eosinophil	Bilobed nucleus with prominent granules	Bilobed nucleus and granules

6. Deep Learning Component

DenseNet121 is used as a pretrained feature-extraction model. The purpose of this stage is to obtain useful visual representations from blood-cell images without training the complete network from the beginning. The extracted features provide information that can support recognition of visual patterns relevant to WBC subtype classification.

7. System Output

The intended output is an annotated blood-cell image. Detected cell-like regions are displayed with their predicted subtype labels, and the system provides the number of detected cells. This output is designed to make the automated result easier to interpret.

8. Advantages

- Automated WBC subtype identification.
- Reduced dependence on completely manual inspection.
- Annotated image output for easier interpretation.
- Use of pretrained deep-learning features.

9. Limitations

- Requires initial setup and processing time.
- Currently trained for four WBC classes only.
- Not trained to classify RBCs or platelets.
- Cell counting depends on the quality of image segmentation.

10. Future Scope

- Add more blood-cell classes.
- Develop separate RBC and platelet classification.
- Improve detection of overlapping cells.
- Use larger and more diverse datasets.
- Develop a user-friendly web or mobile interface.

11. Conclusion

The Blood Cell Subtype Identifier project demonstrates an approach for automating WBC subtype identification from microscopic blood-cell images. By combining image processing with pretrained DenseNet121 features, the system is designed to detect cell-like regions and classify them into four target

subtypes: eosinophils, lymphocytes, monocytes, and neutrophils. The current system is limited to these four classes, and cell-counting performance depends on image segmentation quality. Future development can extend the system to additional blood-cell classes, improve overlapping-cell detection, use larger datasets, and provide a user-friendly interface.

References

[1] Team 19, “Blood Cell Subtype Identifier,” supplied project presentation.

[2] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, “Densely Connected Convolutional Networks,” Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2017.

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